Format-Matched Process Supervision Transfers Better: A Controlled Diagnosis of Supervision Mismatch in Process Reward Model Distillation

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Abstract

Process reward models (PRMs) are commonly trained from automatically generated step-level supervision, but when a small verifier fails it is often unclear whether the bottleneck is student capacity, the training recipe, or the supervision itself. This distinction matters for efficient reasoning systems: if supervision distribution is the bottleneck, then scaling student compute or searching over nearby training recipes is wasteful until the data problem is addressed. We study this question under a controlled setting that holds the student architecture, optimization path, and evaluation benchmark fixed while varying the source and format of process supervision. Using the same Qwen2.5-3B score-head verifier and full ProcessBench-MATH evaluation, ProcessBench-style labels from non-MATH source splits transfer reliably to MATH: GSM8K-source supervision reaches 0.7515 mean ROC-AUC over four seeds, and OmniMath-source supervision reaches 0.7694. In contrast, generated teacher-label supervision is markedly weaker under the same evaluation path: the best generated-label baseline reaches 0.6324 ROC-AUC, and a privileged-teacher-label BCE baseline reaches only 0.5503. Sequence-cluster bootstrap tests over whole MATH solutions show that every GSM8K and OmniMath ProcessBench-source seed significantly beats both generated-label baselines. We further show that the failure is not explained by the score-head machinery alone: a matched ProcessBench-gold diagnostic reaches 0.7129 ROC-AUC, while frozen-representation probes trained on generated teacher labels remain weak. Our comparison does not isolate human-curated versus generated provenance, because provenance, source distribution, and label format vary together. The narrower conclusion is practical: for small PRM distillation, a cheap format-matched diagnostic can reveal supervision mismatch before investing in larger students or broader recipe searches.

1 Introduction

Process reward models assign scores to intermediate reasoning steps and are increasingly used to supervise or verify mathematical reasoning systems. Because human step-level annotation is expensive, many PRM pipelines rely on automatically generated labels from stronger models. This raises a practical question for efficient reasoning systems: when a distilled verifier performs poorly, should we scale the student and training budget, or should we first question whether the generated supervision matches the target evaluation distribution? If supervision distribution is the primary bottleneck, then scaling student compute or searching over nearby training recipes is wasteful until the data problem is addressed.

We investigate this question with a controlled ablation. We fix the verifier architecture, training path, and evaluation benchmark, and vary the source and format of step-level supervision. The student is a Qwen2.5-3B model with a lightweight score head. The target benchmark is full ProcessBench-MATH, evaluated over 1,000 shuffled solutions containing 6,505 reasoning steps and 594 error steps. We compare ProcessBench-style supervision from

non-MATH source splits against generated teacher labels from a privileged/no-ground-truth labeling pipeline.

The same 3B verifier that learns a transferable ProcessBench-MATH scorer from GSM8K or OmniMath ProcessBench-style labels remains weak when trained on generated teacher labels. This contrast is not a state-of-the-art PRM claim: a public Qwen2.5-Math-7B-PRM800K baseline is stronger on the same split. Instead, the contribution is diagnostic. Under a fixed student and fixed optimizer, supervision compatibility accounts for a large difference in transfer quality. This suggests that some failed PRM distillation runs should be treated as data-distribution failures rather than as evidence that the student or recipe is inherently too small.

As a secondary finding, we show that teacher-level privilege does not reliably transfer into student-level verification, traceable to high symmetric label churn between privileged and non-privileged generated labels.

2 Experimental Setup

Student and training path. All main trained verifiers use the same Qwen2.5-3B-Instruct backbone with a scalar score head. We train with LoRA adapters, a score-only objective, balanced batches, binary cross-entropy for the ProcessBench-source runs, and 500 training steps. For generated-label baselines, we report the strongest generated-label checkpoints found in our preliminary ablations: a privileged-teacher BCE run and a no-ground-truth rank-loss run. These generated-label baselines are not presented as a complete optimization sweep; they are included to test whether generated teacher labels can approach the same target under the same evaluation path. The generated-label baselines use one seed each, while the ProcessBench-source verifiers use four seeds. This seed asymmetry is a limitation: additional generated-label seeds could reduce the gap. However, the preliminary generated-label ablations span roughly ten nearby score-head training configurations, and none produces a verifier competitive with the ProcessBench-source transfers (Appendix B).

Evaluation. All headline results evaluate on the same shuffled ProcessBench-MATH split of 1,000 solutions. The split contains 6,505 total steps and 594 error steps. We use exact serial scoring with batch size 1 for headline metrics, because batched score-head evaluation introduced small score perturbations that can affect ranking metrics. We report ROC-AUC and PR-AUC as threshold-free measures. Fixed-threshold F1 is shown only as a diagnostic, because score-head calibration varies across training conditions.

Label sources. ProcessBench-source training uses the first 400 ProcessBench examples from each source configuration, flattened into step-level labels. The GSM8K source contains 2,082 steps, the OmniMath source contains 3,384 steps, and the OlympiadBench source contains 3,579 steps. Exact problem-string overlap between each source file and the MATH1000 evaluation split is zero.

Generated-label training uses automatically labeled MATH-style traces from Gemma-4-26B-A4B, a publicly available mixture-of-experts teacher with roughly 26B total parameters and about 4B active parameters per token. We distinguish labels generated with privileged access to the reference solution from labels generated without ground-truth access. The teacher-level privilege effect is real in isolation, but the student-level transfer question is whether these generated labels yield a better answer-blind verifier. We verified that the teacher did not simply copy ground-truth answers: only 19 of 8,008 teacher-labeled steps showed strong answer leakage (0.24%), confirming that the teacher's labels reflect step-level judgment rather than answer copying.

Uncertainty. Because multiple steps come from the same solution, the primary significance test resamples whole MATH solutions rather than individual steps. We use a paired sequence-cluster bootstrap with 2,000 samples for ROC-AUC gaps. These intervals quantify evaluation-sampling uncertainty for fixed checkpoints; they do not include training-seed variance for the one-seed generated-label baselines.

Training source	Seeds	ROC-AUC	PR-AUC	Best F1	Fixed F1
GSM8K ProcessBench labels → MATH1000	0–3	0.7515 (0.7256–0.7760)	0.2188 (0.1935–0.2317)	0.3144	0.2503
OmniMath ProcessBench labels → MATH1000	0–3	0.7694 (0.7539–0.7869)	0.2524 (0.2277–0.2877)	0.3276	0.3055
Generated privileged teacher labels, BCE	0	0.5503	0.0992	0.1803	0.0000
Generated no-GT teacher labels, rank loss	0	0.6324	0.1418	0.2221	0.2151
Qwen2.5-Math-7B-PRM800K public baseline	0	0.8379	0.3254	0.3991	0.3953

Table 1: Full ProcessBench-MATH transfer results on the same 1,000 MATH solutions. Parentheses show seed ranges. Best F1 is threshold-swept on the evaluation slice and is included only as a diagnostic.

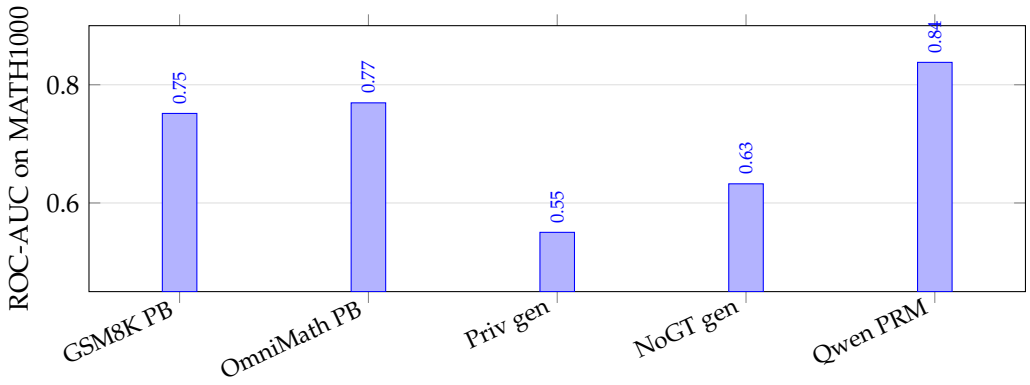


Figure 1: Main transfer pattern: with the same evaluation path, ProcessBench-style source labels transfer much better than generated teacher labels. Qwen PRM800K is an external calibration baseline, not a matched training recipe.

3 Results

3.1 ProcessBench-style source labels transfer; generated teacher labels remain weak

Table 1 shows the main result. With the student, training path, and evaluation held fixed, ProcessBench-style labels from GSM8K and OmniMath transfer strongly to full ProcessBench-MATH. Generated teacher labels are much weaker, even when the generated-label baseline is chosen as the strongest checkpoint from the preliminary ablations.

Figure 1 visualizes the same ranking pattern. The contrast is large enough that it is unlikely to be an artifact of evaluation sampling. Every GSM8K and OmniMath ProcessBench-source seed significantly beats the privileged generated-label BCE baseline under sequence-cluster bootstrap. Against the best generated-label baseline overall, the sequence-cluster ROC-AUC gaps range from +0.0930 to +0.1546 across the eight GSM8K/OmniMath seeds, with two-sided $p = 0.0010$ in all cases. The PR-AUC gaps are proportionally larger, which is important because error steps are sparse: only 594 of 6,505 MATH evaluation steps are labeled as errors.

The public Qwen2.5-Math-7B-PRM800K checkpoint is included as an external calibration point, not as a matched training recipe. It reaches 0.8379 ROC-AUC and remains stronger than any single 500-step ProcessBench-source seed. This confirms that our result should not be framed as state-of-the-art PRM performance. The central observation is instead a supervised-data compatibility mismatch: under the same small-student path, compatible ProcessBench-style supervision transfers while the current generated teacher labels do not.

3.2 The source distribution still matters

The transfer result is not a blanket claim that any non-MATH process labels transfer cleanly to MATH. OlympiadBench is high-variance in our runs: two seeds give 0.5854 and 0.7163 ROC-AUC on full MATH1000. One seed is significantly below the best generated-label

baseline, while the other is significantly above it. We therefore treat OlympiadBench as a boundary diagnostic rather than as a third headline source. The pattern supports a source-distribution view: ProcessBench-style supervision can transfer, but the source split still matters.

3.3 Threshold calibration agrees with the ranking result

Fixed-threshold F1 is fragile for score-head verifiers. To avoid choosing a threshold on the same examples used for reporting, we choose thresholds on the first 200 MATH solutions and evaluate on the remaining 800. Under this held-out calibration protocol, GSM8K ProcessBench-source seeds reach calibrated F1 between 0.2714 and 0.3166, and OmniMath ProcessBench-source seeds reach 0.3062 to 0.3316. The generated privileged BCE baseline reaches 0.1673, and the best generated-label baseline reaches 0.2085. The calibrated-threshold view therefore matches the threshold-free conclusion: ProcessBench-source verifiers remain well above generated-label baselines.

4 Analysis

4.1 The failure is not explained by the score-head machinery alone

A plausible concern is that the 3B score-head architecture is simply too weak to learn step-level mathematical verification. Our diagnostics argue against that as the primary explanation. Using the same score-head path on a matched ProcessBench-MATH split, training on the first 200 shuffled MATH examples and evaluating on the next 200 reaches 0.7129 ROC-AUC. This is not a benchmark claim because train and evaluation data come from the same benchmark family, but it shows that the score-head machinery can learn the target when labels and distribution are aligned (Table 2).

Frozen-representation probes tell the same story. Class-balanced logistic probes on frozen Qwen2.5-3B step-boundary embeddings trained on generated teacher labels transfer weakly to ProcessBench-MATH, around 0.52–0.56 ROC-AUC. The corresponding ProcessBench-gold diagnostic reaches 0.6308 ROC-AUC. This suggests that the failure is not merely that the student representation cannot expose the target; the supervision distribution is doing meaningful work (Table 2). The frozen-probe gap is smaller than the full score-head gap, so we treat it as a supporting diagnostic rather than a standalone explanation.

4.2 Generated privileged labels are not reliably better student supervision

The generated-label results also caution against a simple "better teacher labeler implies better student" story. At the teacher level, privileged access to the worked reference solution improves error detection on MATH. However, at the student level, the privileged labels do not yield a robust verifier advantage. In the N=1000 paired Best-of-N evaluation, neither verifier clears majority vote, and the no-ground-truth verifier is at least competitive with the privileged verifier. This is a separate downstream evaluation measuring solution-level re-ranking accuracy, distinct from the step-level ProcessBench metrics in Table 1. The paired bootstrap on step-level ROC-AUC gives a priv-minus-noGT gap of -0.0098, with a 95% confidence interval of [-0.0252, 0.0053]. We do not interpret the privileged and no-ground-truth rows in Table 1 as an isolated privilege comparison, because they use different score losses. At fixed generated-label objectives, the sign varies across ablations (Table 5); the safe conclusion is that we find no robust privileged-student advantage in this pipeline.

To understand why privileged teacher labels do not yield better student supervision, we examined label-level agreement. Privileged and no-ground-truth labels agree on only 69.1% of steps, indicating that privilege does change the training signal. However, the disagreements are nearly symmetric: neither condition systematically corrects the other's errors. This symmetric churn means the privileged signal does not provide a consistent directional gradient for the student to learn from. In other words, privileged information helps the teacher perform inference-time verification, but the resulting label changes are diffuse rather than corrective.

4.3 Training-recipe changes did not rescue generated labels

We also tested several score-head and distillation variants before reframing the result as a supervision mismatch (Table 5). Naive score-plus-critique distillation was unstable until a feedback-token truncation bug was fixed; after the fix, equal-weight critique distillation remained weak or negative for privileged labels. Soft BCE was mostly vacuous because teacher scores were almost always binary. Pure ranking loss and BCE-plus-rank did not rescue privileged transfer; they often favored the no-ground-truth labels. Longer BCE training increased variance and did not turn the generated privileged labels into a competent MATH verifier. These negative ablations do not prove that no generated-label method can work, but they reduce the likelihood that the observed gap is just a single bad optimizer setting.

5 Related Work

Process reward models for mathematical reasoning. PRMs have been used to provide step-level supervision and test-time verification for mathematical reasoning systems. Math-Shepherd studies automatic process supervision for mathematical reasoning Wang et al. (2023), while Qwen2.5-Math PRMs provide strong public baselines trained at much larger scale Zhang et al. (2025). ProcessBench evaluates earliest-error identification across multiple math reasoning sources and highlights generalization failures of existing reward models Zheng et al. (2024).

Weak and generated process supervision. Recent weak-supervision approaches attempt to train PRMs without dense human process labels, using outcome labels, generated rationales, or model-derived signals Sun et al. (2025). These methods make PRM training more scalable, but they also raise the question studied here: when generated labels fail, is the problem the student, the recipe, or the supervision distribution? Our experiment is complementary: rather than proposing a new weak-supervision method, we hold the small-student path fixed and compare compatible ProcessBench-style process labels against generated teacher labels.

Cross-source and out-of-distribution PRM behavior. Recent and concurrent work studies PRM transfer across GSM8K, MATH, OlympiadBench, OmniMath, and related sources Bai et al. (2026); Zhu et al. (2025). Those studies motivate our source-distribution framing. Our contribution is the controlled contrast between generated teacher labels and ProcessBench-style source labels under the same student, optimizer, and evaluation path.

Data-centric learning and privileged information. Data-centric AI work argues that systematic changes in data quality can dominate model-side changes Mazumder et al. (2022). Our result is a PRM-specific instance of that pattern: a fixed small verifier changes substantially with the supervision source. The privileged-label experiments are also related to learning using privileged information Vapnik and Vashist (2009), but with an important boundary: privileged information can improve the teacher's per-example inference without producing labels that distill into a better answer-blind student.

6 Limitations

This study is diagnostic rather than prescriptive. We evaluate one main student architecture and score-head training path, so a different student family or a substantially different distillation objective may narrow the gap. The main comparison varies several properties at once: ProcessBench-style source labels differ from generated teacher labels in provenance, source problem distribution, and format compatibility with the MATH evaluation. Therefore, this paper should not be read as isolating human-curated versus generated provenance. The missing controlled cell is generated teacher labels on the same GSM8K/OmniMath source problems used by the ProcessBench-style training rows. The generated-label baselines

are the strongest checkpoints from our preliminary ablations, but they are not an exhaustive search over all possible teacher prompting and label-format choices. They also use fewer seeds than the ProcessBench-source runs. We therefore cannot rule out that additional generated-label seeds might reduce the gap, although the breadth of the preliminary ablations makes a pure unlucky-seed explanation less likely. The ProcessBench-source labels share the evaluation benchmark's annotation format, so part of their advantage may come from format compatibility with the evaluation benchmark. Finally, while the public Qwen2.5-Math-7B-PRM800K baseline anchors the scale of the result, we do not compare against every recent PRM baseline under the same data budget. The efficiency claim is also diagnostic rather than a downstream deployment claim: we do not show that the generated-label verifiers beat majority vote in Best-of-N search. Instead, the practical claim is that a small format-matched diagnostic can reveal whether the supervision distribution is likely to support the target verifier before larger student or recipe sweeps.

7 Conclusion

Under a fixed 3B verifier, training path, and ProcessBench-MATH evaluation, compatible ProcessBench-style supervision transfers much better than the generated teacher labels tested here. The result reframes a failed privileged-feedback distillation attempt as a supervision-format and distribution diagnosis. For efficient PRM development, this suggests a practical ordering: before scaling students or re-running the same distillation loop, test whether the supervision distribution itself can support the target verifier. A practical takeaway is to first run a small diagnostic with format-matched labels before investing in a larger student or a new training recipe.

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A Additional Result Details

A.1 Matched-distribution and frozen-probe diagnostics

Table 2 summarizes the diagnostics used to separate score-head or representation capacity from supervision mismatch. The matched ProcessBench score-head diagnostic shows that the training path can learn the target under distribution-matched labels, while the frozen-probe results show that generated teacher labels remain weak even when the representation is held fixed.

Diagnostic	Condition	ROC-AUC
Score head, matched ProcessBench-MATH 200/200	Gold labels	0.7129
Frozen probe, generated teacher labels	Privileged	0.5497
Frozen probe, generated teacher labels	No-GT	0.5559
Frozen probe, ProcessBench-gold 200/200	Gold labels	0.6308

Table 2: Diagnostics showing that the score-head and frozen representation can expose the target when labels are distribution-matched.

A.2 Sequence-cluster bootstrap gaps

Table 3 reports paired ROC-AUC gaps using sequence-cluster bootstrap over whole MATH solutions. Every GSM8K and OmniMath ProcessBench-source seed significantly beats both generated-label baselines.

Model A	Model B	ROC-AUC gap	95% CI
GSM8K seed 0	generated privileged BCE	+0.1932	[0.1644, 0.2208]
GSM8K seed 1	generated privileged BCE	+0.2256	[0.2010, 0.2500]
GSM8K seed 2	generated privileged BCE	+0.2099	[0.1829, 0.2347]
GSM8K seed 3	generated privileged BCE	+0.1749	[0.1494, 0.1995]
OmniMath seed 0	generated privileged BCE	+0.2033	[0.1757, 0.2316]
OmniMath seed 1	generated privileged BCE	+0.2287	[0.1997, 0.2567]
OmniMath seed 2	generated privileged BCE	+0.2072	[0.1773, 0.2362]
OmniMath seed 3	generated privileged BCE	+0.2366	[0.2090, 0.2632]
GSM8K seed 0	best generated-label baseline	+0.1113	[0.0874, 0.1339]
GSM8K seed 1	best generated-label baseline	+0.1436	[0.1248, 0.1627]
GSM8K seed 2	best generated-label baseline	+0.1279	[0.1067, 0.1488]
GSM8K seed 3	best generated-label baseline	+0.0930	[0.0694, 0.1156]
OmniMath seed 0	best generated-label baseline	+0.1214	[0.0964, 0.1458]
OmniMath seed 1	best generated-label baseline	+0.1468	[0.1254, 0.1676]
OmniMath seed 2	best generated-label baseline	+0.1253	[0.1026, 0.1487]
OmniMath seed 3	best generated-label baseline	+0.1546	[0.1305, 0.1773]

Table 3: Sequence-cluster bootstrap ROC-AUC gaps on the MATH1000 evaluation split. All rows have two-sided $p = 0.0010$ with 2,000 bootstrap samples.

A.3 Held-out threshold calibration

Table 4 reports thresholded metrics with thresholds chosen on the first 200 MATH solutions and evaluated on the remaining 800.

A.4 Training-budget and score-averaging diagnostics

Longer training is mixed. Increasing GSM8K from 500 to 1500 steps improves seeds 0 and 1 by +0.0506 and +0.0351 ROC-AUC, respectively, but hurts seed 2 by -0.0622. A

Model	Calibrated F1	Eval ROC-AUC	Eval PR-AUC	Pred error rate
GSM8K ProcessBench seed 0	0.3166	0.7452	0.2339	0.2488
GSM8K ProcessBench seed 1	0.2977	0.7707	0.2152	0.1466
GSM8K ProcessBench seed 2	0.2969	0.7517	0.2261	0.2559
GSM8K ProcessBench seed 3	0.2714	0.7233	0.1940	0.3155
OmniMath ProcessBench seed 0	0.3062	0.7483	0.2292	0.2742
OmniMath ProcessBench seed 1	0.3294	0.7723	0.2606	0.2321
OmniMath ProcessBench seed 2	0.3148	0.7529	0.2327	0.2056
OmniMath ProcessBench seed 3	0.3316	0.7855	0.2818	0.1987
Generated privileged BCE	0.1673	0.5490	0.1002	0.3212
Best generated-label baseline	0.2085	0.6314	0.1431	0.1695
Qwen2.5-Math-7B-PRM800K	0.3855	0.8372	0.3242	0.1091

Table 4: Held-out threshold calibration. Thresholds are chosen on the first 200 MATH solutions and evaluated on the remaining 800.

three-seed mean of the GSM8K 1500-step checkpoints reaches 0.8153 ROC-AUC and 0.3130 PR-AUC, but remains below Qwen2.5-Math-7B-PRM800K under sequence-cluster bootstrap. OmniMath at 1500 steps is high-variance and does not replace the four-seed 500-step main table.

Score averaging is a strong post-hoc diagnostic. Averaging the four GSM8K seeds gives 0.7832 ROC-AUC, averaging the four OmniMath seeds gives 0.7865, and averaging all eight GSM8K+OmniMath seeds gives 0.8073. A mixed seven-member average combining three GSM8K 1500-step seeds with four OmniMath 500-step seeds reaches 0.8276 ROC-AUC and 0.3433 PR-AUC. This is close to the public Qwen baseline by ROC-AUC, but it is an ensemble diagnostic rather than a single-model training claim.

B Generated-Label Ablation Details

The generated-label recipe space included score-plus-critique MSE, score-only training, lower learning rates, reduced language-model loss weight, BCE with error weighting, rank-only loss, BCE-plus-rank, and longer BCE training. The best clean privileged generated-label ROC-AUC among these preliminary ablations was a short BCE run, but it remained weak and did not become a competent verifier. Rank-only and BCE-plus-rank strongly favored no-ground-truth labels. These failures motivated the shift from capacity scaling to the supervision-distribution analysis in the main text.

Generated-label recipe	Priv ROC-AUC	NoGT ROC-AUC	Note
Score+critique MSE, early 3B run	0.4582	0.4870	collapsed near all-negative threshold behavior
Score+critique MSE, batch-size 1	0.5329	0.4857	unstable, many skipped/non-finite examples before truncation fix
Score-only MSE	0.4844	0.4776	stable but weak
Lower learning rate	0.4959	0.4697	collapsed/weak
Feedback-token truncation fixed	0.4514	0.5093	stable but negative for privileged labels
Reduced critique LM weight	0.4546	0.4933	stable but weak
BCE, error weight 3	0.5411	0.4700	best short privileged generated-label ablation, still weak
Rank-only, balanced batches	0.4939	0.6306	noGT wins strongly
BCE+rank, balanced batches	0.5214	0.6140	noGT wins strongly
BCE, error weight 3, longer training	0.4782	0.4971	longer BCE degrades/nulls the privileged signal

Table 5: Preliminary generated-label ablations on the Qwen2.5-3B score-head path. These ablations are not the main result, but they show that the generated-label weakness is not limited to a single score-head recipe.

C Reproducibility and LLM-use statement

The main ProcessBench-source verifier runs use Qwen2.5-3B-Instruct with LoRA rank 16, LoRA alpha 32, dropout 0.05, and all linear layers targeted. The score-head runs use a score-only objective, BCE loss, balanced batches, 500 training steps, batch size 2, model learning rate 10^{-4}, score-head learning rate 5×10^{-5}, weight decay 0.01, and serial ProcessBench evaluation with batch size 1. The source files contain the first 400 ProcessBench examples

from GSM8K or OmniMath, flattened to per-step labels. Generated-label student training is offline: the teacher is not loaded during student training or evaluation because the labeled JSONL files already contain the teacher scores and feedback. The main source-transfer jobs were run as independent single-GPU jobs; downstream Best-of-N generation and scoring were treated separately from the step-level ProcessBench metrics.

LLM assistance was used for manuscript drafting, editing, and formatting. All reported numerical results come from experiment logs and result JSON files produced by the evaluation scripts; no reported metric was generated by an LLM.